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The Rapid Adoption of Generative AI

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1 Introduction

Generative artificial intelligence (genAI) has quickly emerged as a potentially important workplace technology. The large language model ChatGPT debuted in November 2022 and, by December 2024, had over 300 million weekly users. Several studies find that genAI boosts productivity in specific cases (Brynjolfsson et al., 2023; Cui et al., 2024; Dell’Acqua et al., 2023; Demirci et al., 2025; Dillon et al., 2025; Kanazawa et al., 2025; Noy and Zhang, 2023; Otis et al., 2024; Peng et al., 2023), and some project genAI will have large macroeconomic effects (Chui et al., 2023; Filippucci et al., 2024). Yet others predict only modest aggregate impacts (Acemoglu, 2024). These differing views largely reflect uncertainty about the extent of genAI adoption today and in the future.¹

This paper studies the adoption of genAI using data from the first nationally representative U.S. surveys of genAI usage at work and at home. Our data come from the Real-Time Population Survey (RPS), a national online labor market survey of working age adults aged 18-64 that has run since 2020 (Bick and Blandin, 2023). We report the combined results of two surveys fielded in August and November 2024, which collectively included more than 10,000 respondents. These surveys include questions about genAI adoption that we adapt from the Current Population Survey’s Computer and Internet Supplement (CIU), which was introduced in 1984 to document computer (and later, internet) adoption. We also introduce novel questions to measure how intensively individuals use the technology, which tasks genAI assists with, how much time it saves them, and workplace barriers or support for adoption. The resulting microdata will be made publicly available, addressing calls for more research that develops and shares novel data (Ethiraj et al., 2019).

We use this data to address three main questions. First, how many individuals are currently using genAI? Pooling our August and November 2024 surveys, 32% of workers reported using genAI for work: 10% used it every work day, 17% on some but not all work days, and 5% did not use it in the previous week. One third of respondents report using genAI outside of work. Overall, 45% of respondents report using genAI either for work or outside of work. The most commonly used products are ChatGPT (32% of all respondents), Gemini (19%) and embedded products such as Microsoft Copilot (15%).

To benchmark these results, we show that the rate of workplace genAI adoption in 2024, two years after the release of ChatGPT, is similar to the rate of workplace computer adoption in

¹Acemoglu et al. (2022) analyze online job vacancies related to AI, but their data predates ChatGPT. Albanesi et al. (2024), Eisfeldt et al. (2024), and Gathmann et al. (2024) compare occupations according to predictions of AI exposure but do not have data on actual adoption. Humlum and Vestergaard (2025) find no meaningful impacts of genAI adoption on employment and earnings in Denmark among workers in 11 occupations that are particularly exposed to genAI.

1984, three years after the release of the IBM PC (the first mass-market computer). The computer and related information technologies induced far-reaching changes in firm organization and management, which led to substantial productivity gains (Brynjolfsson and Hitt, 1996). The similar pace of genAI adoption thus far suggests that it has the potential to be similarly disruptive. Moreover, we show that genAI adoption by workers is proceeding at a faster pace than adoption by firms, highlighting both challenges and opportunities for managers seeking to capitalize on this new technology (Babina et al., 2024; Bonney et al., 2024; Doshi and Moore, 2025; McElheran et al., 2024).²

Second, which workers are currently using genAI, and what factors contribute to an individual’s adoption decision? We find substantial variation in genAI adoption across demographic and labor market characteristics. Adoption has been faster for male, younger, more educated, and higher-wage workers. Adoption also differs greatly by occupation, suggesting that differences in job roles and responsibilities are an important determinant of adoption. Consistent with this, we show that recent estimates of genAI task-based exposure are highly predictive of actual adoption (Eloundou et al., 2024; Felten et al., 2021), and that occupations with lower adoption also have higher shares of workers who report that genAI is not useful for their job. These findings contribute to our understanding of how genAI maps onto specific tasks and roles within the firm, which is relevant both for allocating adoption support and adapting firm organization.

We provide evidence that firm climate and policies also appear to play an important role in workers’ adoption choice. A sizable share of workers who do not use genAI cite workplace barriers as a key reason, and this share is particularly large in a few occupations who may deal with sensitive data. Alternatively, workers whose employers encourage them to use genAI are much more likely to do so, even after controlling for other characteristics. This suggests that when job tasks are amenable to genAI adoption, firms can play an active role by promoting usage, offering training, and removing organizational barriers to adoption.

Third, how large are potential productivity gains from genAI? Based on self-reported time savings due to genAI, the average genAI user would need to work 5.2% more hours in order to complete their work without genAI, implying 1.4% time savings across all workers (including non-users). We use a standard production model to estimate potential productivity gains at the industry and aggregate level. We find wide variation in potential productivity gains by industry,

²The Census BTOS Survey estimates between August and November 2024 5.4% of firms had incorporated AI technologies into their production process. This suggests that firm-based measures may miss an important share of genAI adoption. Aaron Levie, the founder and CEO of Box, a cloud storage provider, compares this pattern with past technologies: “It’s almost a one-to-one parallel with the early wave of consumerization of IT. We saw that exact same dynamic in the early 2010s, which was employees of enterprises were pretty frustrated with their traditional document management and collaboration technology, and so they would go out and just go onto a web browser, sign up for a Box account, and then bring it into their organization.” (Levie, 2024)

Non-employed respondents are not shown the term in brackets. While the CIU asks about computer use “at home”, we use slightly different language to account for respondents who use genAI on mobile devices outside of their home.

The survey asks several follow-up questions of genAI users regarding which products they used, how intensively they used it, and how much time they saved, always distinguishing between work and non-work use.

An important caveat regarding our measurement approach is that it will only capture genAI use that respondents are aware of. For example, while typing prompts into ChatGPT is an obvious case of using genAI, automatic AI-assisted writing suggestions are a more subtle case that respondents may not consider. Because our estimates may not detect some passive or embedded uses of the technology, we interpret our estimates a lower bound on the extent of genAI adoption.

3 How Prevalent is Generative AI Use?

Figure 1a displays the share of August and November 2024 RPS respondents who report using genAI. The samples for the “Overall” and “Outside of Work” bars are all respondents, while the sample for the “For Work” bar is employed respondents. The overall estimates reflect the share who use genAI either for work or outside of work.

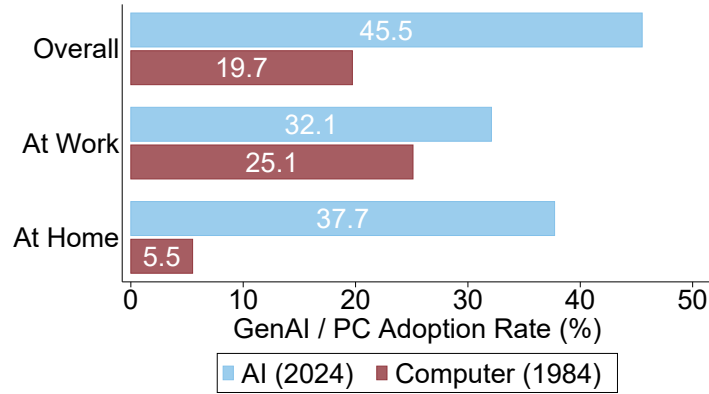
The first bar shows that 45.5% of respondents report using genAI either at work or at home: 10.4% used it every day, 26.7% used it some but not all days, and 8.3% did not use it in the last week. The second bar shows that 32.1% of workers used genAI at work: 10.5% used it every work day and 16.7% on some but not all workdays, bringing total weekly usage to 27.2%. Outside of work, usage is higher overall (37.7%) but less intensive, with only 5.6% of all respondents using it every day last week.

3.1 How Do Our Results Compare to Other Estimates?

The RPS is administered by Qualtrics. To provide a second source of data, in December 2024 we fielded our genAI module through the Survey of Working Arrangements and Attitudes (SWAA) by Barrero et al. (2021).⁵ The SWAA uses a different survey provider (IncQuery) but targets a similar sample intended to be representative of the U.S. population ages 20-64. Figure 1b displays analogous results from the SWAA. 46.5% of SWAA respondents reported using genAI overall, versus 45.5% of RPS respondents. We find similar estimates for work (33.3% SWAA,

⁵See Appendix C for more details on the SWAA, how we construct weights to maximize comparability between the SWAA and RPS, and some validation exercises.

Figure 2: Computer and Generative AI Adoption 2-3 Years After First Mass-Market Product



Notes: The figure displays usage rates for computers (PCs) and genAI in 1984 and 2024, respectively. The IBM PC was released in 1981, which we refer to as the first mass-market computer, implying that 1984 is three years after the first mass-market computer. ChatGPT was released in November 2022, which we refer to as the first mass-market genAI product, implying the fall of 2024 is two years after the first mass-market product for that technology. The data source for genAI is the August and November 2024 wave of the RPS (blue bars). The data source for computers is the 1984 Computer and Internet Use Supplement of the CPS (red bars). The sample for the Overall and At Home bars in the RPS and CPS is all individuals ages 18-64 (RPS, $N = 9726$; CPS, $N = 61,708$). The sample for the For Work bars is employed individuals ages 18-64 (RPS, $N = 6935$; CPS, $N = 89,271$).

measured using very similar questions for both technologies.) The first mass-market computer was the IBM PC, which was released in August 1981. The first mass-market genAI product was ChatGPT, which was released in November 2022.⁸ For the PC, we measure adoption in the CPS CIU for the first time in 1984, i.e. three years after the first mass-market computer. The RPS provides adoption estimates for 2024, two years after the first mass-market genAI product.

The top pair of bars in Figure 2 refers to overall adoption for the two technologies two to three years after the introduction of the first mass-market product. The bar for the RPS repeats the 45% overall adoption rate for genAI reported in Figure 1a. Overall PC adoption is roughly 20% in 1984, implying that genAI has been adopted at a faster pace than the PC. As another more recent comparison, in 1995 the internet was opened to commercial traffic after the decommissioning of NSFNet (Leiner et al., 2009).⁹ Two years later, 21% of US adults reported access to the internet according to the International Telecommunication Union, which compiles subscriber data from national regulatory bodies and service providers (Pena-Lopez et al., 2009). This is a similar pace of overall adoption to PCs, and substantially slower than the adoption of genAI.

⁸Three notable computer products had already been released in 1977 (the Apple II, the Commodore PET, and the TRS-80). Choosing earlier introduction dates would imply slower adoption rates for a given technology. The 1981 IBM PC was the first computer to sell over one million units (Abbate, 1999). We chose these dates based in part on ChatGPT’s answers to questions about the year of the first mass market product for each technology. The exact prompts and answers are available upon request.

⁹1995 was also the year of Netscape’s initial public offering and the year that AOL 3.0 was released.

A key question is whether genAI’s faster adoption was driven by work or non-work use.¹⁰ The faster overall pace of genAI adoption compared with PCs is driven by non-work adoption. The middle pair of bars in Figure 2 plots work adoption of genAI and PCs. We find similar work adoption rates for genAI after two years (32%) and PCs after three years (25%). By contrast, the bottom pair of bars in Figure 2 reveals much higher non-work adoption rates for genAI (38%) compared with PCs (5%).

The computer and related information technologies induced far-reaching changes in firm organization and management, which led to substantial productivity gains (Brynjolfsson and Hitt, 1996). The similar pace of genAI adoption by workers thus far suggests that this new technology could also be transformative for firms. At the same time, the choice of whether to adopt a technology depends on both its benefits and costs. PCs required purchasing expensive immobile hardware and nontrivial training. In contrast, many genAI products are free or inexpensive and user-friendly. On the one hand, this lower adoption cost likely contributes to genAI’s rapid uptake. On the other hand, this means that similar adoption rates for PCs and genAI at work do not necessarily imply equivalent benefits.

4 Which Workers Use Generative AI?

In this section, we document that early genAI adoption has not been uniform throughout the workforce. Understanding which workers are adopting genAI at higher rates provides several important insights. First, because many workers are adopting genAI ahead of firm-level adoption, variation in genAI use can help managers allocate support, training, and integration efforts. Second, we show that employer encouragement of genAI use is highly predictive of adoption, suggesting that firms and managers can influence the nature and extent of genAI use in their organization. Third, adoption patterns shed light on how genAI maps onto specific tasks and roles within the firm, which is relevant for decisions regarding job redesign and firm reorganization.

Figure 3 documents demographic heterogeneity in work adoption of genAI. The top panel shows genAI adoption is 6 percentage points higher for men. The middle panel show genAI adoption by age. Workers aged 18-49 have similar adoption rates, while adoption is lower for workers aged 50-64. The bottom panel shows that adoption rates are increasing in educational attainment. In particular, workers with a Bachelor’s degree or more use genAI at roughly twice the rate of non-college educated workers. Differences between demographic groups are primarily driven by users who use genAI every workday or at least once in the previous week.

Figure 4 plots the kernel density of genAI adoption by weekly earnings percentile. Adoption

¹⁰Unfortunately, internet adoption data do not allow us to distinguish work from non-work adoption.